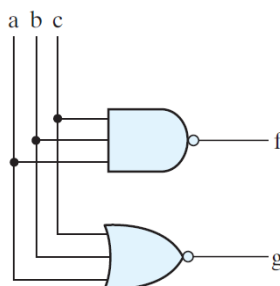


DSM tutorial quiz – 1

- 1) (a) Convert $(0.513)_{10}$ to octal. (b) Convert $(345)_6$ (345 to the base six) to decimal.
(Ans: (a) $(231.406517)_8$ (b) $(137)_{10}$)
- 2) (a) Perform the multiplication: $(1011)_2 \times (110)_2$ (b) Perform the division: $(101101)_2 \div (11)_2$
(Ans: (a) $(1000010)_2$ (b) $(1111)_2$)
- 3) Convert the hexadecimal number 64CD to binary.
(Ans: $(0110010011001101)_2$)
- 4) (a) What is 2's complement used for? Why is 2's complement preferred over 1's complement for signed binary representation, and how do you find the 2's complement of a binary number?
 (b) How to recognize positive and negative numbers in binary?
(Ans: 2's complement is used for representing signed numbers in binary. 2's complement is preferred over 1's complement because it has only one representation for zero and makes addition and subtraction easier to perform. To find the 2's complement: Invert all bits ($0 \rightarrow 1$, $1 \rightarrow 0$) and then add 1.
If the MSB of a signed binary number is 0, it is positive. If MSB is 1, it is negative.)
- 5) What is the largest binary number that can be expressed with 16 bits? What are the equivalent decimal and hexadecimal numbers?
(Ans: $(1111111111111111)_2$, $(65535)_{10}$, $(FFFF)_{16}$)
- 6) Obtain the 1's and 2's complements of the following binary numbers: (a) 10010101
 (b) 00000000
(Ans: (a) 01101010, 01101011 (b) 11111111, 00000000)
- 7) Perform the subtraction $7 - 3$ using 1's complement subtraction.
(Ans: $7=0111$, $3=0011$
1's complement of 3: 1100
Add: $0111+1100=1\ 0011$
End-around carry present $\Rightarrow$ result is positive $\Rightarrow$ add it: $0011+1=0100 \Rightarrow 4$)
- 8) Perform the subtraction $3 - 7$ using 2's complement subtraction.
(Ans: Take the 2's complement of 7: $0111 \rightarrow 1000 \rightarrow 1001$
Add it to 3: $0011+1001=1100$. There is no carry, so the result is negative.
Take the 2's complement of 1100 to find its magnitude: $1100 \rightarrow 0011 \rightarrow 0100$
 $\Rightarrow$ Result is -4)
- 9) Represent the decimal number 6,248 in BCD.
(Ans: 0110 0010 0100 1000)
- 10) Give the truth tables for f and g:



Ans: f -> NAND, g -> NOR

a	b	c	f	g
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0	0	0	1	1
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	0	0